

Deep inelastic scattering on polarized lithium: the physics behind the planned measurements

2026-08-17 · companion to the cos 2φ projection report and to plans/01 (physics case) · written for students and colleagues joining the program; external numbers carry over from the verified physics-case survey

Polarized lithium beams would provide the EIC with spin-polarized *nuclei* that are light enough for ab initio computation yet complex enough to exhibit genuine nuclear structure. ⁶Li is a spin-1 nucleus well described as a deuteron bound to an inert α core; ⁷Li is its spin-3/2 partner, an α plus a triton. This note develops, starting from the DIS formalism, the four measurements around which the program is built: the tensor structure function b₁, the double-helicity-flip ("exotic glue") structure function Δ, the polarized EMC effect, and spectator-tagged DIS on the embedded deuteron.

1 • Polarized lithium at the EIC

The EIC baseline includes polarized electrons, protons, deuterons, and ³He. The EPIOS white paper [1] presents the accelerator case for adding polarized ⁶Li and ⁷Li: an atomic-beam source with optical pumping at 671 nm feeding EBIS charge breeding, with source targets P_z ≥ 0.90 (vector) and P_{zz} ≥ 0.80 (tensor) at ≥ 10¹³ atoms/s. In the ring, ⁶Li is spin-dynamically similar to the deuteron: its small gyromagnetic anomaly (G = −0.178) implies only 81 linear depolarizing resonances up to the top energy of ~138 GeV/u, compared with 2457 for ³He, so the machine techniques developed for deuterons carry over. ⁷Li (G = +1.532) reaches ~117 GeV/u with partial snakes and tune jumps [1].

Four measurements motivate the beam, each addressing a verified gap in the literature: no b₁ prediction or measurement exists for any nucleus beyond the deuteron; no gluon-transversity projection exists for any EIC target; no study of α-tagged DIS on lithium exists; and no lithium-beam luminosity or polarization parameter set has been published. The purpose of the program's simulations is to convert these gaps into figures of merit.

MEASUREMENT	OBSERVABLE	ELECTRON BEAM	ION SPIN	AXIS
Polarized EMC effect (⁷ Li, ⁶ Li)	A → g ₁ ^A	polarized, helicity flipped	vector P _z	longitudinal
Tensor structure function (⁶ Li)	A _{zz} → b ₁	unpolarized	tensor P _{zz}	longitudinal (or along q)
Exotic glue (⁶ Li; ⁷ Li eligible)	cos 2φ → Δ(x, Q ²)	unpolarized	tensor P _{zz}	transverse
Tagged DIS (both isotopes)	tagged versions of all three	as above	as above	as above, correlated with spectator

2 • Deep inelastic scattering

An electron with four-momentum k scatters from a nucleus of momentum P by exchanging a virtual photon q = k − k'. The kinematic invariants, in the per-nucleon convention used throughout the program (energies and structure functions per nucleon, x ≤ 1), are

$$Q^2 = -q^2, \quad x = \frac{Q^2}{2P \cdot q}, \quad y = \frac{P \cdot q}{P \cdot k}, \quad Q^2 = x y s, \quad W^2 = M^2 + Q^2 \frac{1-x}{x}$$

where x is the momentum fraction of the struck parton, y the energy-transfer fraction, and W the invariant mass of the hadronic final state. The unpolarized cross section is carried by two structure functions,

$$\frac{d^2\sigma}{dx dQ^2} = \frac{4\pi\alpha^2}{xQ^4} \left[\left(1 - y + \frac{y^2}{2}\right) F_2(x, Q^2) - \frac{y^2}{2} F_L(x, Q^2) \right], \quad F_1 = \frac{F_2}{2x(1+R)}$$

with R = σ_L/σ_T. In the parton model, F₂ = Σ_q e²_q x[q(x)+q̄(x)]: the cross section measures quark distributions weighted by squared charge. For a nucleus we use F₂^A/A, so rates are quoted against luminosity per nucleon.

Polarization introduces additional structure functions beyond F_1 and F_2 ; their number and character depend on the target spin.

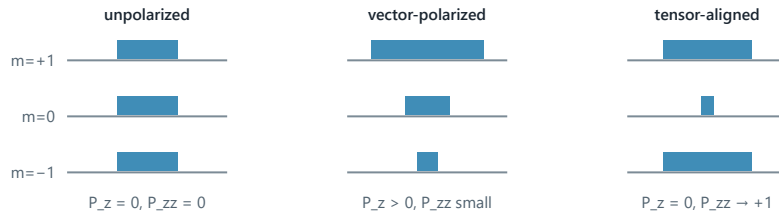
3 • Spin-1 targets

A spin- $1/2$ target contributes two spin structure functions, g_1 and g_2 . The principal observable is g_1 , the helicity-weighted quark distribution, measured through the double-spin asymmetry $A_{//} \approx D \cdot g_1 / F_1$ (D is a kinematic depolarization factor) with both beams longitudinally polarized and the electron helicity reversed bunch by bunch.

A spin-1 target carries additional structure. Its spin state along a chosen axis is specified by three populations p_+ , p_0 , p_- of the projections $m = +1, 0, -1$, and two moments characterize the fill:

$$P_z = p_+ - p_- \in [-1, 1], \quad P_{zz} = p_+ + p_- - 2p_0 = 1 - 3p_0 \in [-2, 1]$$

P_z is the familiar vector polarization. P_{zz} , the *tensor* or alignment polarization, is specific to spin ≥ 1 : it distinguishes spin along $\pm z$ from spin in the transverse plane ($m = 0$), and it is nonzero even when equal numbers point up and down. An unpolarized beam has populations ($1/3, 1/3, 1/3$) and both moments vanish; a pure $m = 0$ beam has $P_z = 0$ but $P_{zz} = -2$.



Hoodbhoy, Jaffe, and Manohar showed [2] that a spin-1 target has, beyond F_1 , F_2 , g_1 , g_2 , four tensor-polarized structure functions b_1 – b_4 , and Jaffe and Manohar identified one more, the double-helicity-flip function Δ [3]. At leading twist the tensor sector reduces to b_1 (with $b_2 = 2x b_1$ as its Callan–Gross partner) and Δ . For unpolarized electrons on a target whose spin axis makes angle θ_m with the beam, the cross section for each pure m state is [2,3]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1) \text{ for } m = (+1, 0, -1)$$

The weights (1, -2, 1) organize all tensor observables: the $m = \pm 1$ states enter together and against $m = 0$, precisely the combination measured by P_{zz} . Averaging the formula over a fill with populations p_m replaces every c_m by a factor P_{zz} . Two experimental handles follow. With the spin axis *longitudinal*, the rate difference between fills gives $A_{zz} \propto b_1$; with the spin axis *transverse* ($\theta_m = 90^\circ$), the ϕ distribution of the scattered electron acquires a $\cos 2\phi$ modulation proportional to Δ . Both require tensor polarization; neither requires polarized electrons.

^7Li , with spin $3/2$, has four populations and three moments (vector, tensor/quadrupole, and octupole). Its rank-0 and rank-1 sectors are exact analogs of the spin- $1/2$ case, and host the polarized EMC measurement; its rank-2 sector is the formal analog of the spin-1 tensor sector, for which no structure-function basis has been published — an open theory item the program tracks.

4 • The measurement program

4.1 • The tensor structure function b_1

In parton language, b_1 compares quark distributions in an aligned versus a transverse nucleus [2]:

$$b_1(x) = \frac{1}{2} \sum_q e_q^2 \left[q^{(0)}(x) - \frac{q^{(+1)}(x) + q^{(-1)}(x)}{2} \right], \quad \int_0^1 b_1(x) dx = 0 \quad (\text{Close – Kumano, no tensor sea})$$

where $q^{(m)}$ is the quark distribution in a target with spin projection m . For a collection of free spin-1/2 nucleons, or for a pure S-wave bound state, all m states are equivalent and $b_1 = 0$. A nonzero b_1 therefore measures directly how binding modifies the partonic content: D-wave admixture, pion exchange, hidden-color components, or a tensor-polarized sea (which would violate the Close–Kumano sum rule [4]).

The only existing measurement is HERMES on the deuteron [5]: A_{zz} up to 2×10^{-2} over $0.01 < x < 0.45$ and $b_1(x = 0.012) = 0.112 \pm 0.055 \pm 0.028$, unexpectedly large at low x , with a zero crossing near $x \approx 0.2$ and a sum-rule integral 2σ from zero. Standard convolution calculations predict $|b_1| < 10^{-3}$ at $x \gtrsim 0.2$, an order of magnitude below the data [6], whereas a model with pions and a hidden-color probability of only 0.15% reproduces it [7]. The tension is unresolved; JLab E12-13-011 will remeasure the high- x region, and the EIC adds the low- x region, where HERMES observed the large signal, together with the Q^2 lever arm.

Measurement procedure. The beam is delivered in fills of different tensor polarization ("tensor thirds"). With the estimator $(N_+ + N_- - 2N_0)$ over matched luminosities, all m -independent physics cancels, and the statistical uncertainty on A_{zz} is $\sqrt{(2/N)/P_{zz}}$. ${}^6\text{Li}$ would provide the first b_1 of any $A > 2$ nucleus, for which no prediction exists, and the difference between ${}^6\text{Li}$ and the free deuteron isolates the effect of embedding in the α field. One systematic is tracked from the outset: at collider kinematics a clean b_1 extraction favors orienting the spin axis along the momentum transfer $\mathbf{q}$ rather than along the beam [26].

4.2 • $\Delta(x, Q^2)$: exotic glue

The $\cos 2\phi$ term in the master formula corresponds to the virtual photon flipping its helicity by two units between absorption and emission in the forward Compton amplitude. A spin-1/2 particle cannot absorb two units of helicity, so neither quarks nor bound nucleons generate Δ at leading twist; gluons can. Δ is the structure function that the gluon transversity distribution, an interference of gluon helicity states existing only in targets with $J \geq 1$, generates at order α_s through a photon–gluon convolution [3,9]. A nonzero Δ in a nucleus is therefore direct evidence for glue not inherited from individual nucleons, hence the name "exotic glue". It has not been measured for any target; the one prior experimental document is a JLab letter of intent [13].

The measurement uses unpolarized electrons on a tensor-polarized nucleus whose alignment axis lies in the transverse plane, and fits the azimuthal distribution about that axis. Averaged over the fill, the modulation amplitude and its inversion back to Δ are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2$$

The expected size spans per-mille to percent. The only absolute normalizations available are moment estimates: a bag-model calculation by Sather and Schmidt for the spin-3/2 Δ^{++} baryon, $\int x \Delta dx = -0.012 \alpha_s$ [9], and lattice-QCD calculations by Detmold and Shanahan of gluon transversity in the ϕ meson (clearly nonzero, albeit at an unphysical pion mass) [10], with NPLQCD finding the analogous matrix element roughly ten times suppressed in the deuteron [11]. Percent-level asymmetries appear only in proton–deuteron Drell–Yan under the maximal assumption $\Delta_{Tg} = \Delta_g$, described by its authors as optimistic [12]. No calculation exists for ${}^6\text{Li}$ or any $A > 2$ nucleus, and the literature brackets the nuclear Δ in both directions: suppressed, as in the lattice deuteron, or enhanced by binding in analogy with the EMC effect [1]. The projection report transfers the bag moment to ${}^6\text{Li}$ as a labeled scenario, shows that one EIC year measures the amplitude at $\geq 20\sigma$ per bin, and details the caveats.

The coherent channel. The same modulation can be measured in coherent diffraction, $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$, in which the nucleus remains in its ground state and recoils intact. There the $\cos 2\phi$ modulation mixes two mechanisms: the nuclear *shape*, in which a deformed gluon density modulates the diffractive slope (the mechanism computed for polarized deuterons in the only existing calculation of this kind [23]), and exotic glue proper. ${}^6\text{Li}$ discriminates between the two because its quadrupole moment is anomalously small ($Q = -0.0806 \text{ fm}^2$, roughly 50 times smaller than that of ${}^7\text{Li}$): the shape term is bounded and linear in t , so a flat-in- t

modulation cannot be a shape effect. Detection of the intact recoil is itself nontrivial: the nucleus retains the beam rigidity exactly ($A/Z = 2$) and emerges within ~ 2 mrad of the beam, visible only in the Roman-Pot transverse-momentum tail at IP6 or at the secondary focus of a second interaction region [25,27]; the companion report treats this in detail.

4.3 • The polarized EMC effect: g_1 of a bound proton

The EMC effect — the $\sim 10\text{--}20\%$ suppression of F_2 per nucleon in a nucleus at $0.3 \lesssim x \lesssim 0.7$ — has been measured for forty years without a settled explanation. Its spin analog has never been measured at all: does the *spin* structure of a bound nucleon change more, less, or the same as its momentum structure? The observable is the polarized EMC ratio

$$\Delta R_A(x) = \frac{g_1^A(x)}{P_p g_1^p(x) + P_n g_1^n(x)}$$

where P_p, P_n are the effective nucleon polarizations from nuclear structure. ${}^7\text{Li}$ is well suited: quantum Monte Carlo gives $P_p = +0.866$ and $P_n = -0.037$ [17], so to a good approximation polarized ${}^7\text{Li}$ is one polarized proton plus six spectators, with computable corrections. Two classes of models make sharply different predictions: Cloët–Bentz–Thomas (mean-field, NJL) obtain a polarized EMC effect about twice the unpolarized one [14], whereas Tronchin–Matevosyan–Thomas obtain equal effects [15]. A measurement at the 5% level discriminates between them, and the two map onto the competing mean-field and short-range-correlation pictures of the EMC effect itself. A handle unique to the EIC is the Q^2 evolution of g_1 , driven by the polarized gluon distribution; the predicted *gluon-spin* EMC effect is larger still, with ${}^7\text{Li}$ identified as the most promising case [16]. JLab E12-14-001 will measure ${}^7\text{Li}$ at $x > 0.06$; the EIC extends the coverage to $x < 0.06$ and adds the Q^2 arm, in collider kinematics free of target-fragment absorption.

For this observable both beams are polarized longitudinally and the electron helicity flips bunch-by-bunch; the error scales as $1/(P_e P_Z \sqrt{N})$, so it uses the vector polarization the source delivers most readily.

4.4 • Spectator-tagged DIS

The cluster description of ${}^6\text{Li}$ is quantitative: cluster-model calculations give the deuteron-like pair a polarization of ≈ 0.87 within ${}^6\text{Li}$, carrying over 90% of the nuclear spin [18]. When the α spectator is detected (it carries beam velocity into the far-forward spectrometers), the struck object is identified as the deuteron cluster, and the ${}^6\text{Li}$ beam acts as a source of polarized deuterons at controlled virtuality. The formal framework is spectator tagging (Cosyn–Weiss [20]): the spectator momentum is measured and extrapolated in its transverse component to the free-nucleon (here free-cluster) pole, recovering free-particle structure with binding effects removed by construction rather than modeled. Tagged *tensor* asymmetries are predicted to reach $O(1)$ at spectator momenta of ~ 300 MeV/c, where the D-wave dominates — far larger than the inclusive asymmetries above. The benchmark for the method is the $e+{}^3\text{He}$ double-tagging study of Frišćić et al. [21], in which tagging improved on effective-polarization unfolding by a factor of 2–3, and by a factor of 10 at low x .

Far-forward routing determines where each isotope’s tag fragment is detected, and it follows from magnetic rigidity alone: a beam-velocity fragment with mass-to-charge ratio A/Z bends in the hadron optics according to $R = (A/Z)_{\text{frag}}/(A/Z)_{\text{beam}}$.

BEAM	TAG FRAGMENT	R	DESTINATION AT IP6
${}^7\text{Li}$ ($A/Z = 7/3$)	α	0.857	Roman Pots main window ($R \in 0.60\text{--}0.95$) — works now
${}^7\text{Li}$	t	1.286	over-rigid, no coverage
${}^6\text{Li}$ ($A/Z = 2$)	α or d	1.000	beam rigidity: inside the beam envelope except the p_T tail
${}^6\text{Li}$	p / n	0.50 / –	Off-Momentum Detectors / ZDC (veto handles)

The tagging program therefore begins with ${}^7\text{Li}$ at IP6, while ${}^6\text{Li}$ α -tagging (and the coherent intact- ${}^6\text{Li}$ channel, which faces the same rigidity problem) motivates the secondary focus of the second interaction region [27]. This

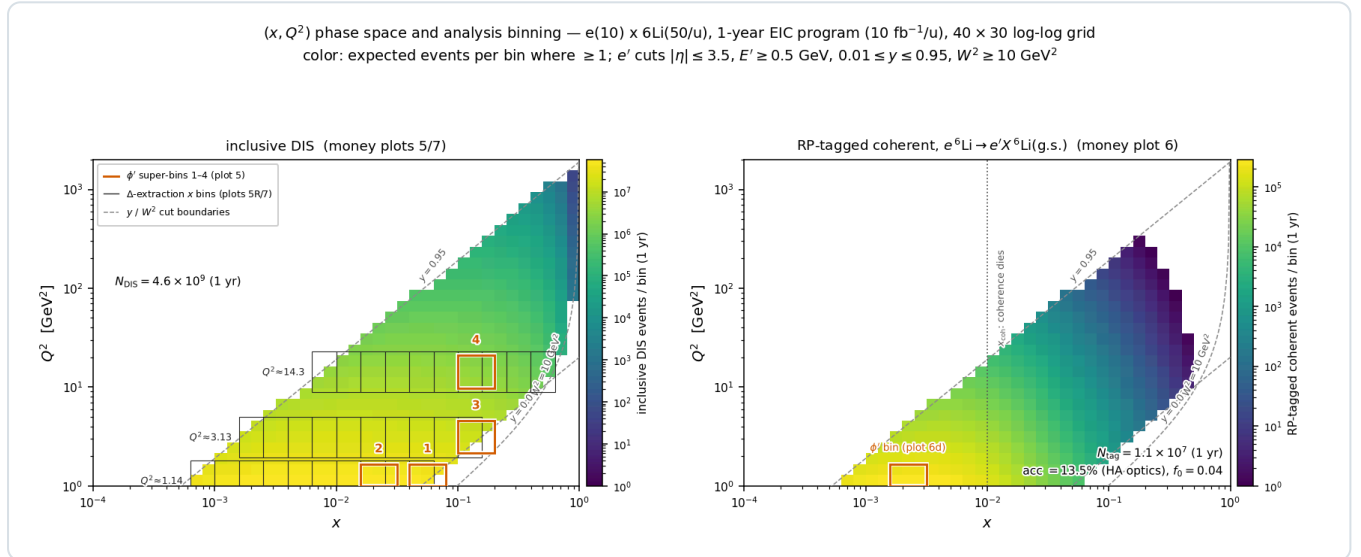
asymmetry between the isotopes was itself a finding of the simulation program.

5 • Rates and statistical precision

All projections use luminosity per nucleon, with $10 \text{ fb}^{-1}/u$ corresponding to approximately one year and $100 \text{ fb}^{-1}/u$ to ten years at nominal EIC rates, and in-ring polarization placeholders $P_z = 0.7$ and $P_{zz} = 0.6$ (ring transport from the source values is an open accelerator question). The statistical uncertainties behind every money plot are

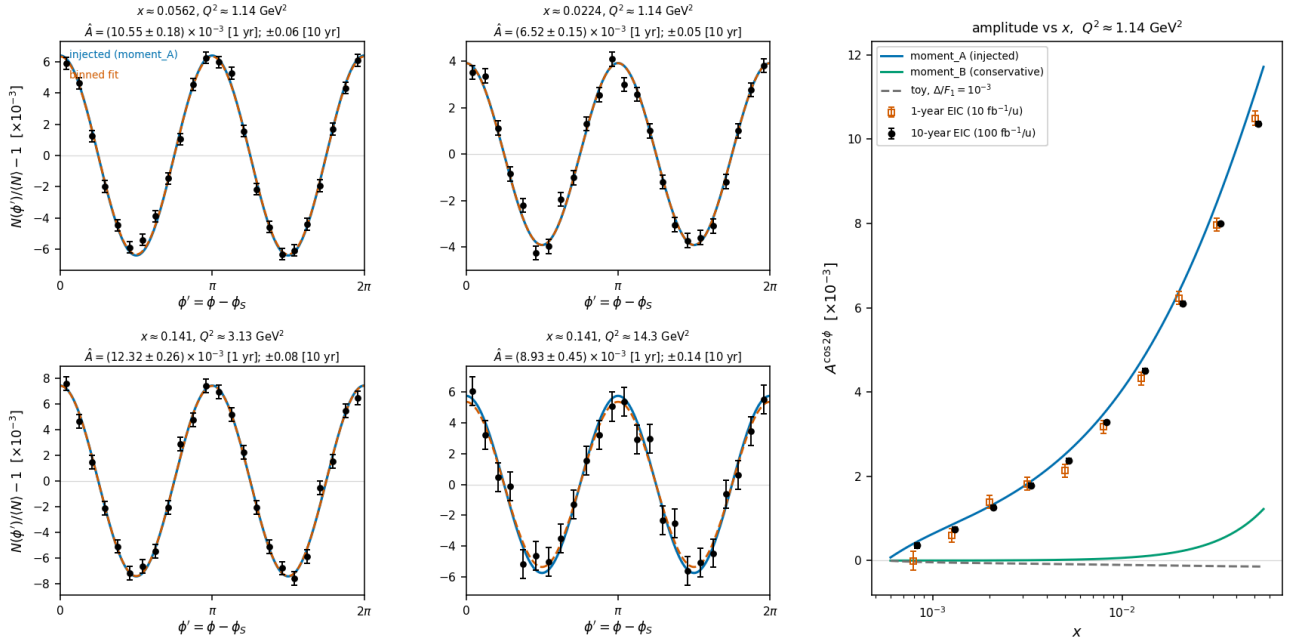
$$\delta A_{\parallel} = \frac{1}{P_e P_z \sqrt{N}}, \quad \delta A_{zz} = \frac{\sqrt{2/N}}{P_{zz}}, \quad \delta A^{\cos 2\phi} = \frac{\sqrt{2/N}}{P_{zz}}$$

The factor $\sqrt{2}$ in the tensor uncertainties follows from $\langle \cos^2 2\phi \rangle = 1/2$ (equivalently, from the three-fill estimator variance); the polarization appears in the denominator because the fill dilutes the physical asymmetry. These formulas, and their behavior under realistic ϕ -acceptance holes (where a binned fit replaces the naive moment), are validated end to end by pseudo-experiments with the polligen event generator. Tensor observables enjoy a structural advantage in systematics: the $\cos 2\phi$ modulation is self-normalized within a single fill, so relative-luminosity errors largely cancel; what remains is the polarimetry scale ($\delta P_{zz}/P_{zz} \approx 3\%$) and radiative corrections, which for tensor observables have not yet been calculated.



Phase-space coverage. Expected events per (x, Q^2) bin in one EIC year at $e(10 \text{ GeV}) \times {}^6\text{Li}(50 \text{ GeV}/u)$. Left: inclusive DIS (4.6×10^9 accepted events) with the bins of the $\cos 2\phi$ analysis overlaid; the measurement is limited by polarization, acceptance, and systematics rather than by statistics. Right: the coherent intact- ${}^6\text{Li}$ channel (1.1×10^7 tagged events), confined to $x \lesssim 0.03$ because coherence requires the photon to resolve the whole nucleus, a condition that fails once $x \gtrsim 0.01$.

Nuclear gluonometry, transversely tensor-polarized ${}^6\text{Li}$, $e(10) \times 6\text{Li}(50/u)$, $P_{zz} = 0.60$
moment_A(variant=mid_x, c=-0.012, A=-0.2941, q2_ref=3.913, dilution=0.3333); ϕ' at 1 yr (10 fb $^{-1}/u$), amplitudes 1 & 10 yr — stat. only; bkg & tensor RC unquantified (plans/0)



Projected extraction. $\cos 2\phi'$ pseudo-data (one EIC year, $P_{zz} = 0.6$) in the four highest-significance (x, Q^2) bins, with the injected moment-constrained Δ model and the binned fit; right, the extracted amplitude versus x against alternative model readings. The projected precision distinguishes between the models rather than only from zero. Assumptions and caveats are detailed in the companion report.

6 • Further reading

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Educational companion to [reports/cos2phi_money_plots_report.html](#) (projections and caveats) and [plans/01_findings_physics_case.md](#) (sourced physics case). Generated by [reports/build_report.py](#); display math typeset at build time (matplotlib mathtext, STIX), figures from the evgen scripts. Projections use scenario structure functions with explicit bands.